



OFFICIAL PRODUCT DOCUMENTATION

Pocket Ledger V2

The definitive local-first expense tracker for Android. Complete sovereignty over your personal financial records with instant offline SQLite performance and user-controlled Google Sheets cloud synchronization.

✓ Local Room SQLite Primary

✓ Foreground-Only Sync

✓ Zero Background Battery Drain

✓ Private Google Sheets Backup

✓ Interactive Analytics & Widget

100% Local-First Speed

Every expense commits instantly to the device's Room database. The interface refreshes immediately with zero loading spinners or network latency.

Foreground Batch Sync

Cloud sync pushes only when triggered by saving an expense or tapping Sync Now. All pending items upload in a single high-efficiency atomic batch.

Visual Spending Analytics

Gain instant clarity with custom Canvas donut charts, daily expenditure trend lines, multi-period filtering, and comprehensive category breakdowns.

Ergonomic Quick Access

Log expenses in seconds using the responsive home screen widget or direct in-app quick entry.

**Application
Package**

com.pocketledger.v2

Target Platform

Android 8.0+ (API 26–36)

Primary StorageRoom SQLite
(pocket_ledger_v2.db)**Cloud Backup Mirror**Google Sheets via Apps
Script**Developer &
Support**

Namyia / @TechyBist

● SYSTEM ARCHITECTURE

The Local-First Philosophy

Why Pocket Ledger V2 is fundamentally engineered for privacy, instant speed, and explicit cloud control.

TRADITIONAL CLOUD EXPENSE TRACKERS	POCKET LEDGER V2 (LOCAL-FIRST)
Cloud Database is Primary: If offline, app hangs or throws errors.	Device SQLite is Primary: 100% functional offline; zero internet needed.
Network Latency: Must wait 1–3s for remote API confirmation.	Instant UI Updates: Transactions write locally; UI reflects immediately.
Background Battery Drain: Background polling, alarms, wake-locks.	Zero Background Execution: No background services, workers, or timers.
Third-Party Server Custody: Personal finances sit in vendor databases.	Complete Data Ownership: Data resides on phone & your personal Google Drive.

⚡ Architectural Pipeline



The Four Non-Negotiable Sync Rules

- 1. Local-First Guarantee:** Every transaction is committed to the on-device Room SQLite database first. The app will never fail to record an expense because of a weak cellular signal.
- 2. Explicit Foreground Execution:** Cloud synchronization runs strictly while the app is active in the foreground—either immediately after saving an expense or when you tap "Sync Now". Pocket Ledger contains zero background services, zero periodic alarms, and zero scheduled sync tasks.
- 3. Resilient Offline Safety:** If network connectivity drops or the Google Apps Script endpoint is temporarily unreachable, the transaction remains safely stored locally in Room DB with its pending status preserved. It will automatically upload during the next foreground sync.
- 4. Soft-Delete Tombstones & Single Batch:** Deletions write local tombstone records so the cloud mirror reflects removals cleanly. All pending creates, updates, and deletes are transmitted in a single atomic batch payload to eliminate cold-start latency.



Complete Independence from Google Sheets

You are never forced to set up Google Sheets. Pocket Ledger functions as a 100% complete, fully featured personal finance tracker completely offline. Google Sheets acts strictly as your private cloud backup.

• GETTING STARTED

Initial Setup & Cloud Pairing

Set up your local application and link your personal Google Sheet in four straightforward steps.

1 Install the Pocket Ledger Android App

Obtain the release APK (**app-release.apk**) from the distribution directory or project website. Install it on any device running Android 8.0 (API 26) or newer. Grant notification permissions if prompted (Android 13+) to enable local confirmation banners.

2 Create Your Private V2 Google Sheet Copy

Open the official Android V2 Google Sheet template copy link in your web browser:

```
https://docs.google.com/spreadsheets/d/1_h4vU5tBAL690L78D-TeeV8beGiSub9cUTMVFmmiPcI/copy
```

Click **"Make a copy"**. Google Drive duplicates the template directly into your account with all required columns A–L, data formulas, and monthly summary aggregations intact.

3 Deploy the Apps Script Web App

In your newly created Google Sheet, navigate to **Extensions → Apps Script** from the top menu bar.

- Click the blue **Deploy → New deployment** button at top right.
- In the modal, click the gear icon next to "Select type" and choose **Web app**.
- Set Description: **Pocket Ledger V2**
- Set Execute as: **Me (your email address)**
- Set Who has access: **Anyone** (required for your Android app to post)
- Click **Deploy**, grant Google account access permissions, and copy the resulting **Web app URL**.

4 Pair URL in Pocket Ledger Settings

Launch Pocket Ledger on your phone. Tap the **Settings** icon in the bottom navigation bar → select **Sync & Google Sheets**. Paste your copied Web App URL into the text box and tap **Test Connection**. Once verified, tap **Sync Now**.

Google Sheet Backend Schema Architecture (Columns A–L)

Your V2 Google Sheet template mirrors the Room database with 12 structured columns:

Col A: Date (D/M/YYYY)

Col B: Month (Full Name)

Col C: Time (HH:MM)

Col D: Category (With Emoji)

Col E: Amount (Numeric)

Col F: Payment Mode (Name)

Col G: Remarks (Context)

Col H: Device Name (Tag)

Col I: Transaction ID (UUID)

Col J: Created At (Epoch)

Col K: Updated At (Epoch)

Col L: Deleted (0 or 1)



Strict Platform Isolation — Android V2 vs iPhone

The copy link above and Apps Script V2 backend are strictly for Android V2. **Do NOT connect an iPhone Shortcut to this V2 sheet.** iPhone remains on its independent V1 sheet and legacy Apps Script backend.

● SPENDING ANALYTICS

The Home Dashboard

Real-time spending aggregation, dynamic visual charts, and instant period filtering.

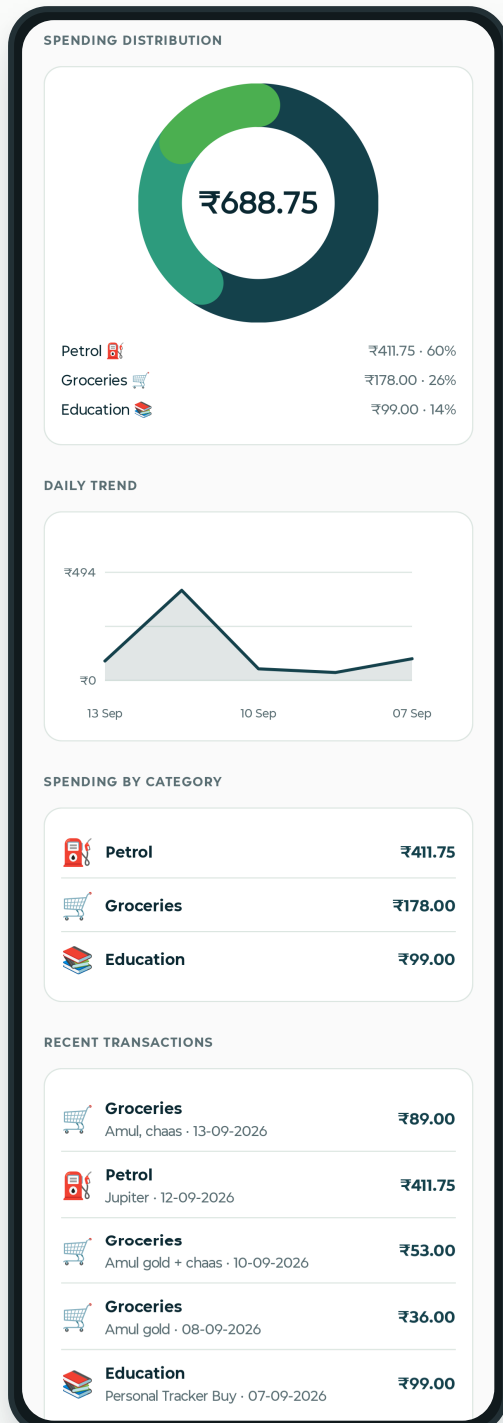


Figure 3.1: V2 Comprehensive Home Dashboard

Interactive Period Selector

Filter all spending analytics on the fly with four discrete time ranges: **Today**, **This Week**, **This Month**, and **Last Month**. All calculations occur locally in Room DB with zero query delay.

Key Performance Indicator (KPI) Cards

High-contrast summary metric cards deliver immediate financial clarity:

- **Total Expenses:** Cumulative expenditure for the selected period.
- **Avg per Expense:** Mathematical mean spend per logged transaction.

Custom Donut Distribution Chart

A custom Android Canvas-rendered donut chart illustrates category spending allocations with vibrant color coding and proportional arc slices.

Daily Spending Trend Graph

A bespoke daily expenditure graph visualizes peaks and spending patterns across the chosen timeframe, highlighting high-outlay days at a glance.

Category Breakdown & Recents

Ranked category spending list with emoji badges, transaction counts, exact totals, and percentage shares, followed by an immediate view of recent entries.



Zero Network Spinners

Dashboard analytics are computed directly via SQLite Room aggregations on device—no loading delays ever.

EXPENSE LOGGING

Adding Expenses

Fast, ergonomic bottom-sheet entry designed for rapid single-handed operation.

The screenshot shows the 'Add an expense' bottom sheet. At the top, there's a title 'Add an expense' and a 'Cancel' button. Below it, the 'AMOUNT' field shows '₹ 0.00'. The 'CATEGORY' dropdown is set to 'Other' with a coin icon. The 'PAYMENT MODE' dropdown is set to 'Fampay' with a card icon. A 'NOTE - OPTIONAL' field contains the text 'What was this for?'. Below the note, it says 'Saved to your device. Sync to Google Sheets when ready.' At the bottom of the form is a 'Review & save' button. Below the form is a numeric keypad with digits 1-9, 0, a decimal point, and a minus sign, along with some utility icons like a calculator icon, a close icon, and a globe icon.

Figure 4.1: Quick-Add Expense Bottom Sheet

Quick Entry Methods

Access the bottom-anchored quick-entry sheet instantly:

- 1. Dashboard Action:** Tap the prominent "+ Add Expense" button at the bottom of the home dashboard.
- 2. Home Screen Widget:** Tap the "+" action directly on your placed home screen widget for immediate entry.

Field Anatomy & Validation

- **Date:** Automatically set to current date; tap to open calendar picker.
- **Category:** Dropdown with preconfigured or custom emoji categories.
- **Amount:** Clean numeric pad with currency prefix formatting.
- **Payment Mode:** Cash, UPI, Credit Card, Debit Card, or custom.
- **Remarks:** Optional contextual note (vendor name, receipt notes).

Confirm & Save Architecture

Tapping "Confirm & Save" executes the local-first save flow:

- 1.** Generates a stable UUID and writes the transaction directly to Room SQLite.
- 2.** The sheet immediately closes and the dashboard reflects updated totals.
- 3.** The app initiates a non-blocking foreground sync to Google Sheets.



Double-Submission Protection

The Save button automatically disables upon initial tap to prevent accidental double-billing or duplicate transaction records.

• COMPLETE HISTORY

The Ledger

Search, filter, inspect, and audit your complete financial transaction ledger.

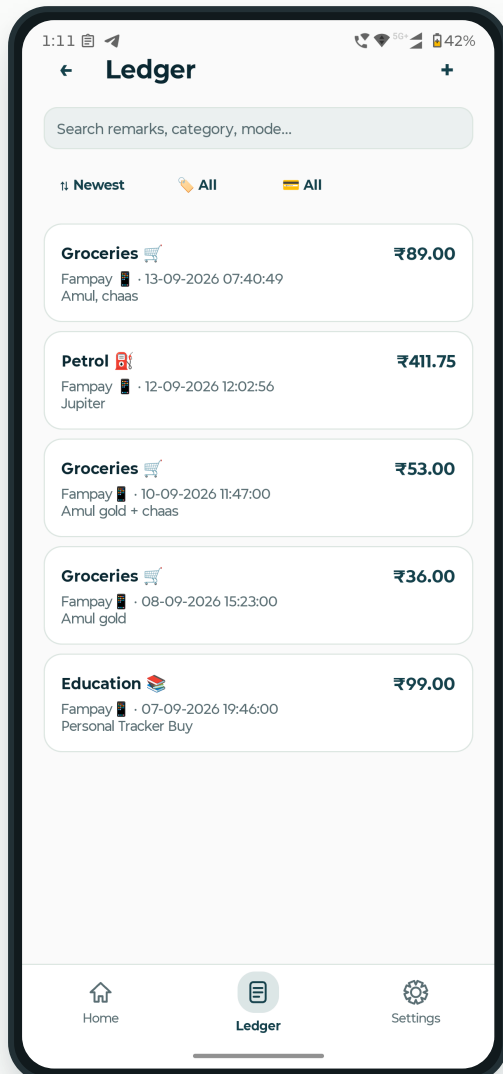


Figure 5.1: Transaction Ledger with Active Filters

Real-Time Search Bar

Type any vendor name, keyword, or note to instantly filter entries. Room SQLite full-text querying scans remarks, categories, and amounts as you type with zero input latency.

Multi-Dimension Filter Controls

Combine multiple filter parameters to pinpoint specific expenditures:

- › **Category Filter:** Narrow down to Food, Groceries, Fuel, Bills, etc.
- › **Payment Mode Filter:** Inspect Cash vs. UPI vs. Card expenses.
- › **Clear Filters:** One-tap reset to view the entire master ledger.

Flexible Sort Options

Sort your financial records ascending or descending:

- › **Date (Newest → Oldest):** Chronological audit order (default).
- › **Date (Oldest → Newest):** Historical sequence review.
- › **Amount (Highest → Lowest):** Identify largest outlays.
- › **Category Name:** Alphabetical category grouping.

Transaction Item Anatomy

Each transaction row renders the category emoji, clean remarks, payment mode tag, exact timestamp, formatted amount, and sync status state.



Interactive Gestures

Long-press any transaction card in the ledger to reveal the action menu for editing or deleting. See Chapter 06 for details.

• MODIFICATIONS

Editing & Deleting Expenses

Correct logging mistakes or remove records with soft-delete cloud sync protection.

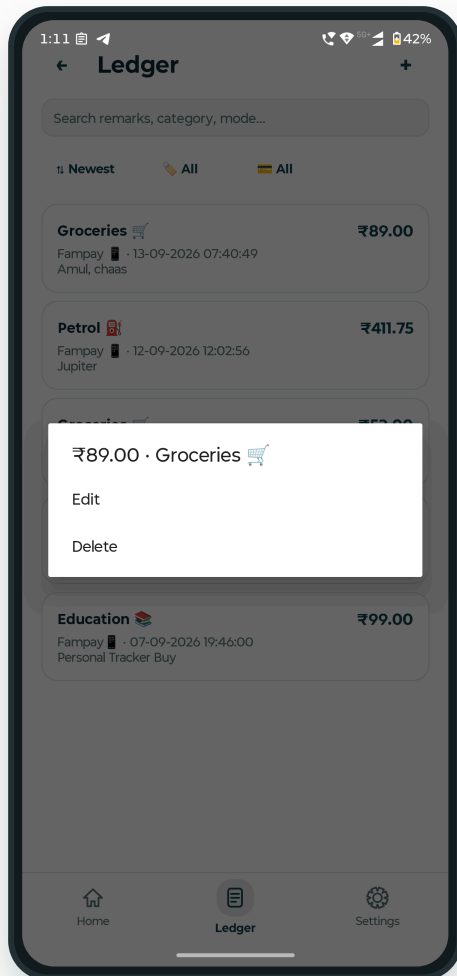


Figure 6.1: Long-Press Action Sheet

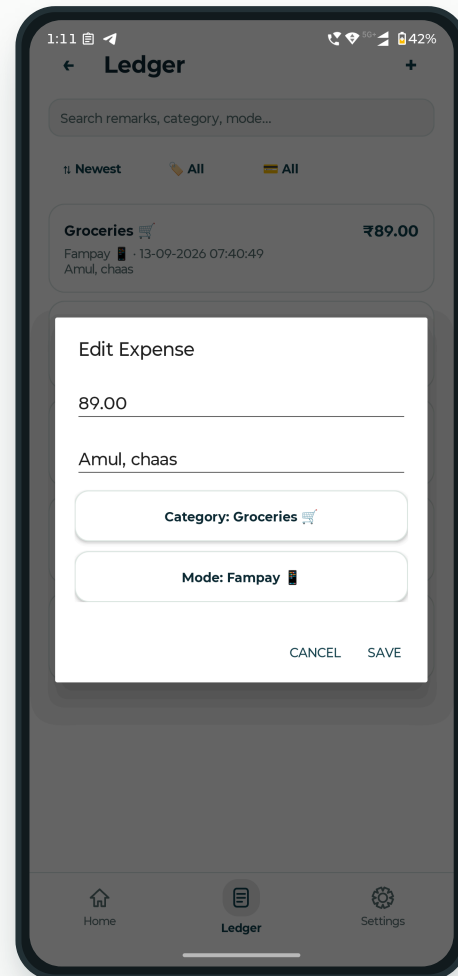


Figure 6.2: Edit Expense Interface

The Long-Press Gesture

In the Ledger view, **touch and hold (long-press)** any transaction row. A contextual bottom sheet emerges containing the selected transaction details along with two dedicated actions: **Edit** and **Delete**.

Editing a Transaction

Tap **"Edit"** in the menu to open the pre-populated Edit Expense screen (Figure 6.2). You can modify:

- › Date, Category, and Amount
- › Payment Mode and Remarks

Tap **"Save Changes"** to write updates to Room DB. The record is flagged for foreground cloud update via action:

UPDATE

Deleting a Transaction

Tap **"Delete"** to open a safety confirmation dialog. Upon confirmation:

- › Record marked as a soft tombstone (**deleted = 1**).
- › Immediately hidden from Ledger & Dashboard.
- › Transmitted to Sheets via action: **DELETE**.

Soft-Delete Tombstone Architecture

Pocket Ledger does not immediately purge deleted rows from the local database before syncing. It stores an internal tombstone so that during the next foreground sync, the Apps Script backend locates the exact row in Google Sheets and removes it cleanly.

● CONFIGURATION

Settings & Google Sheets Sync

Configure your private Apps Script endpoint and manage cloud synchronization.

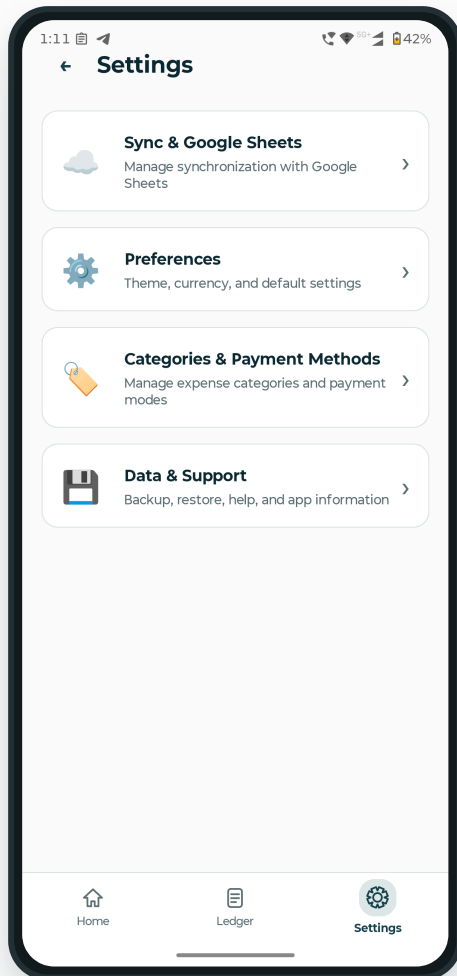


Figure 7.1: Settings Hub

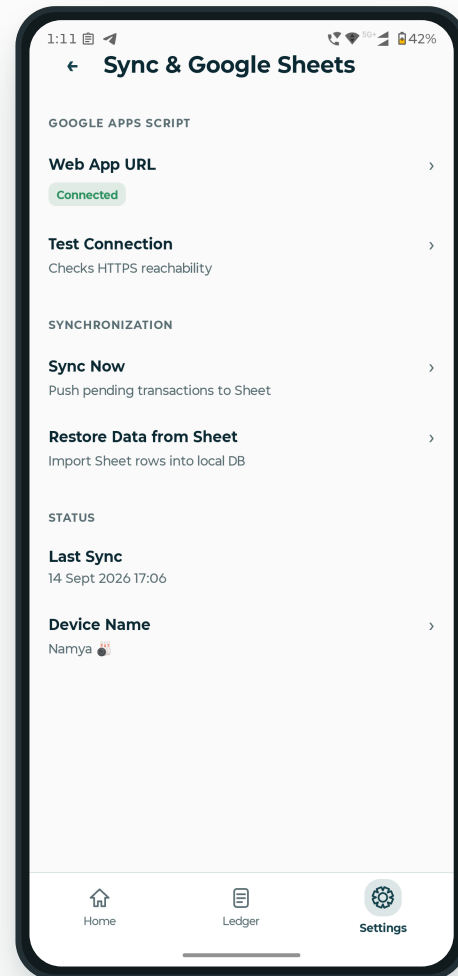


Figure 7.2: Sync & Google Sheets

Sync Controls & Operations

- › **Web App URL:** Stores your private Google Apps Script endpoint.
- › **Test Connection:** Non-mutating GET ping to verify URL health.
- › **Sync Now:** Explicitly triggers immediate batch sync of all pending changes.
- › **Restore Data:** Imports all cloud transactions into local Room DB.

Device & Metadata Info

- › **Device Name:** Identifier attached to each sheet entry (e.g. "Pixel 8").
- › **Last Sync:** Exact timestamp of the last confirmed cloud update.
- › **Batch Protocol:** Uploads pending entries in a single HTTP request.

⚡ Sync Engine Protocol — Explicit & Foreground Only

Sync occurs **only** when triggered by saving an expense while the app is in the foreground, or when you explicitly tap "Sync Now". Pocket Ledger never runs background jobs, polls in the background, or drains device battery when closed.

● PREFERENCES

Preferences & Themes

Customize application appearance, currency formatting, and default entry options.

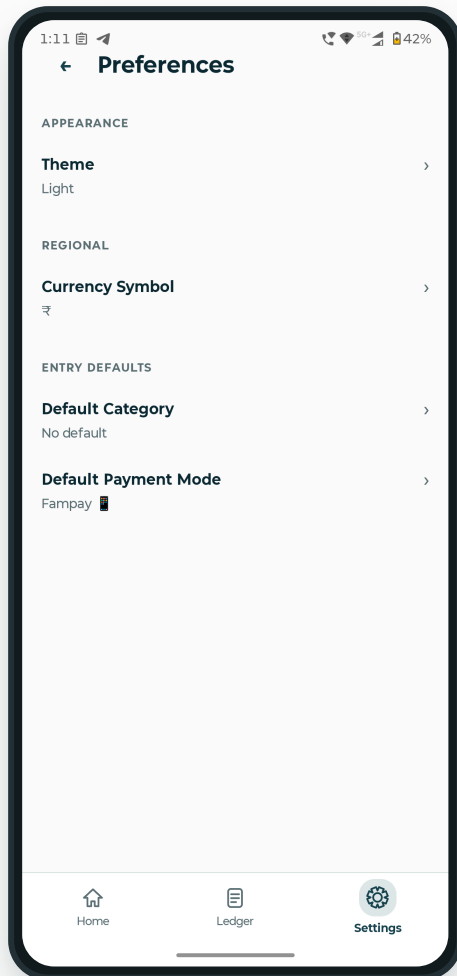


Figure 8.1: Preferences (Light Mode)

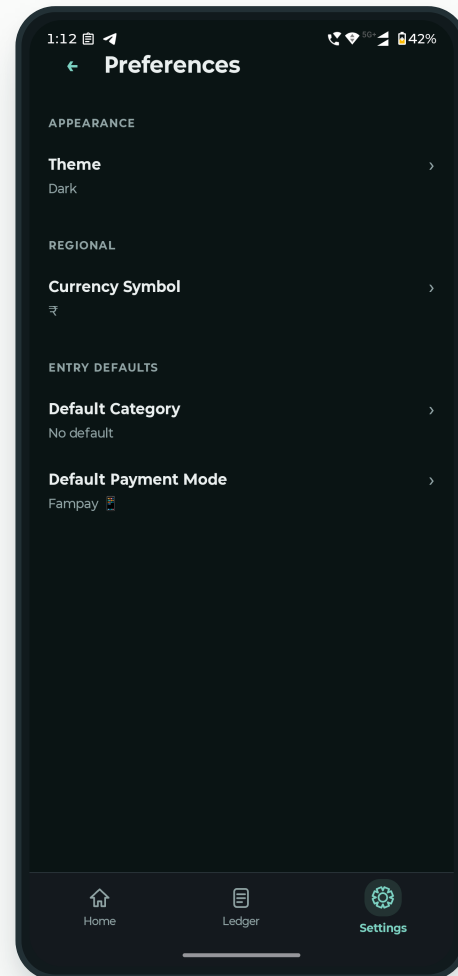


Figure 8.2: Preferences (Dark Mode)

Theme Engine

Select between three display modes:

- › **Light Theme:** High-clarity off-white surface with deep teal accents (#16414F).
- › **Dark Theme:** Deep slate (#0E1416) with radiant mint accents (#7FD6C4) for low-light environments.
- › **System Default:** Dynamically adapts to Android day/night schedule.

Currency & Entry Defaults

Tailor quick-entry defaults to your habits:

- › **Currency Symbol:** Configure ₹, \$, €, £, ¥, or custom symbols across the entire UI.
- › **Default Category:** Pre-select your most frequent category (e.g. Food & Dining).
- › **Default Payment Mode:** Pre-select your go-to payment method (e.g. UPI or Card).



Instant Preference Persistence

Preferences are stored securely in Android SharedPreferences and apply immediately without requiring an application restart.

● PERSONALIZATION

Categories & Payment Methods

Organize your expenses with custom labels, emoji icons, and personalized sort order.

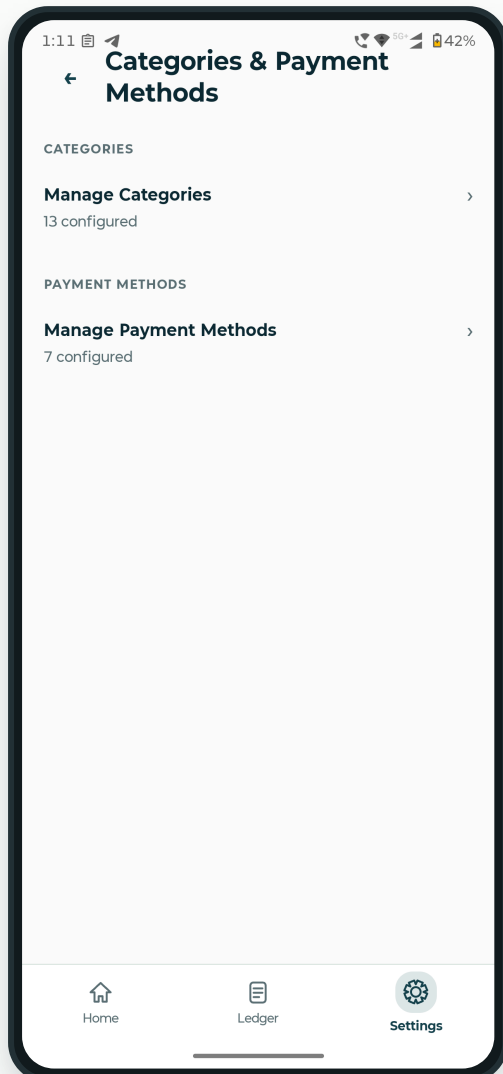


Figure 9.1: Category & Payment Manager

Manage Categories

Personalize your spending buckets:

- › **Add Category:** Tap "+ Add Category", type a label, and assign an emoji (e.g. "Coffee ☕").
- › **Edit & Rename:** Tap any category to modify its title or emoji icon.
- › **Reorder:** Drag reorder handles to place high-frequency categories at the top of dropdowns.
- › **Delete:** Tap delete icon. Deleting locally does not alter historical transactions.

Manage Payment Methods

Track how you pay across all financial accounts:

- › Configure Cash 💵, UPI 🇮🇳, Credit Card 💳, Debit Card 🏠, Net Banking 🌐, etc.
- › Add bank-specific tags (e.g. "HDFC Card 💳" or "Apple Pay 🍏").

Preloaded Defaults (Preserved with Emojis)

Categories: Food 🍔, Groceries 🛒, Transport 🚗, Bills 💡, Shopping 🛒, Health 🏥

Payment Modes: Cash 💵, UPI 🇮🇳, Credit Card 💳, Debit Card 🏠, Net Banking 🌐

💡 Full UTF-8 Emoji Preservation

Pocket Ledger V2 natively preserves emoji glyphs in categories and payment modes. They sync to Google Sheets without ASCII corruption or stripping.

● SECURITY & BACKUP

Data Management & Support

Offline JSON backup/restore, text export, diagnostics, and developer assistance.

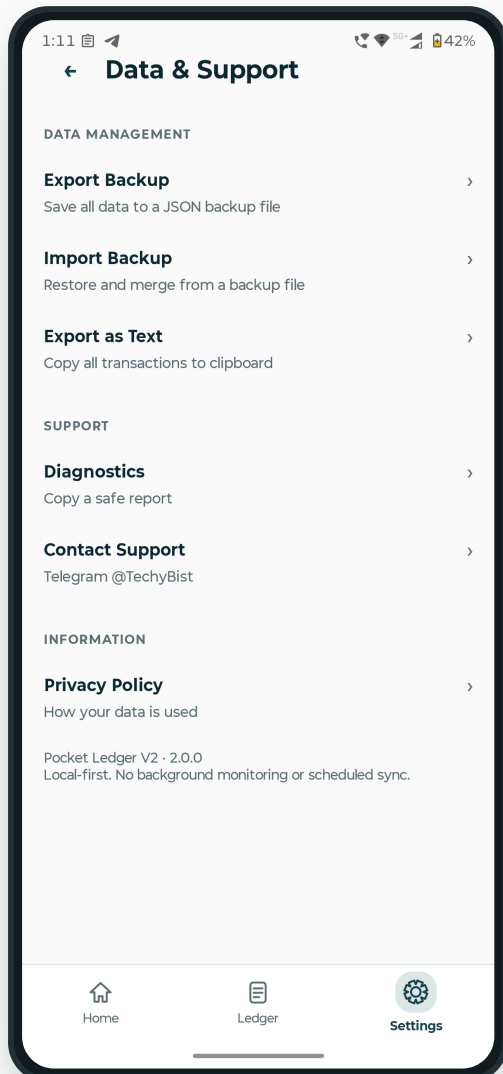


Figure 10.1: Data Management & Support Hub

JSON Export & Import Backup

- › **Export Backup:** Exports an unencrypted, complete JSON backup of all Room DB records to your device's Documents or Downloads folder.
- › **Import Backup:** Restores transactions from any exported JSON backup file on any device running Pocket Ledger V2.

Export as Plain Text

Generates a clean, formatted plain-text log of your transactions suitable for copy-pasting into emails, notes apps, or personal audit records.

Diagnostics & System Logs

View real-time diagnostic telemetry: active database record count, pending sync queue length, device model details, and recent sync HTTP status codes.

Support & Community

Direct access to developer support:

- › **Telegram Support:** [@TechyBist](#)
- › **GitHub Repository:** [Techyeye98/Pocket-Ledger](#)



Offline Privacy Architecture

Pocket Ledger operates directly on your device with no third-party tracking, advertising SDKs, or external account requirements. Your financial data stays strictly between your phone and your personal Google Sheet.

• WIDGETS

Home Screen Widget Setup & Resizing

Instant 1-tap expense logging directly from your Android launcher home screen.

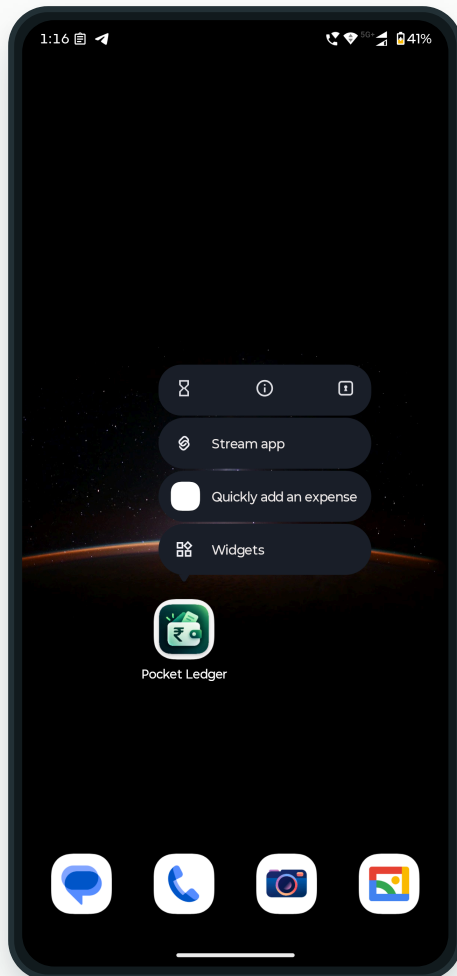


Figure 11.1: Adding Widget via Launcher

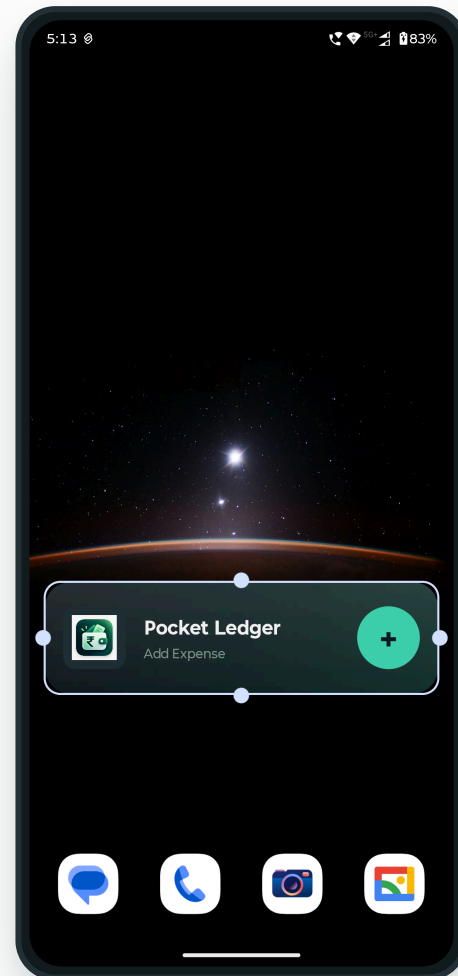


Figure 11.2: Interactive Widget Resizing

Step-by-Step Widget Placement

1. **Long-press** the Pocket Ledger app icon on your home screen (or hold any empty home screen area).
2. Tap **"Widgets"** in the popup menu.
3. Locate Pocket Ledger and drag the **"Add Expense"** widget onto your home screen grid.

Resizing & Layout Adaptation

1. **Long-press** the placed widget until the grid bounding box and resize handles appear.
2. Drag handles outward or inward to expand or contract the widget.
3. Dynamically adapts to widget width across 3 responsive layouts: **Compact** (<160dp), **Standard** (160–279dp), and **Expanded** (≥280dp).



Direct Quick-Entry Trigger

Tapping the "+" icon on the widget launches the quick-entry bottom sheet directly without opening the main dashboard window, letting you log an expense in under three seconds.

● COMPANION & FAQ

iPhone Setup & Troubleshooting FAQ

Independent iOS Shortcut workflow and answers to frequently asked technical questions.

iPhone Companion Architecture (Completely Separate)

Pocket Ledger V2 is a native Android application. For Apple iOS users, an independent Apple Shortcuts workflow is provided:

- **Workflow Pipeline:** iOS Shortcut → Existing Apps Script → Existing V1 Google Sheet.
- **CRITICAL RULE:** Do NOT tell iPhone users to use the V2 Sheet or V2 backend. iPhone remains strictly on its proven V1 architecture and dedicated spreadsheet. Keep the two platforms completely decoupled.

Frequently Asked Questions (FAQ)

Q: Where did my expense go after saving?

A: It was saved immediately to your local device database (Room SQLite). The dashboard and ledger reflect it in real time. If connected to Google Sheets, the app attempts a foreground sync while active.

Q: Sync failed or I was offline. Is my transaction lost?

A: Never. Local Room DB is your primary source of truth. The transaction remains safely stored on your phone with a pending flag and will automatically push during your next foreground sync or when you tap "Sync Now".

Q: The "Test Connection" succeeds, but saving/syncing fails?

A: "Test Connection" checks basic HTTPS GET reachability. For POST transactions, confirm your Apps Script deployment is configured with "Execute as: Me" and "Who has access: Anyone", and ensure no spreadsheet columns were renamed.

Q: Can I use both Android V2 and iPhone on the same Google Sheet?

A: No. Android V2 uses a revised schema with columns A-L (UUIDs, timestamps, tombstones) and batched JSON. iPhone shortcuts use the V1 schema. Always keep them on their respective sheets.

Q: How do I transfer my expense data to a new Android device?

A: Either export an offline JSON backup (Settings → Data & Support → Export Backup) and import it on your new phone, OR configure your Google Sheets Web App URL on the new device and tap "Restore Data from Sheet".

Q: Does Pocket Ledger consume battery in the background?

A: Zero. Pocket Ledger has no background workers, polling threads, or scheduled wake-locks. It consumes resources only when you actively use it.

**Need Technical Assistance?**

Reach out to developer support directly via Telegram at [@TechyBist](https://t.me/TechyBist) or visit the official repository at github.com/Techyeye98/Pocket-Ledger.